AADM: An Architecture Artifact Decision Model for Risk-Aware Documentation Scope

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Abstract

Architecture decision making is a central activity in software engineering processes, yet it is often treated implicitly or reduced to isolated technical choices (van Heesch et al., 2012; Kruchten et al., 2019). This paper proposes an Architecture Decision Model (ADM) that makes architectural decision logic explicit as a structured, repeatable process, addressing a recurring need for decision transparency in evolving software-intensive systems (Bosch, 2014; Hilliard et al., 2020). The model supports process tailoring by allowing decision concerns, decision criteria, and decision artifacts to be systematically adapted to project-specific contexts, reflecting empirical observations that architecture effort must be proportional to project risk and complexity (Waterman et al., 2015). Rather than prescribing a fixed set of architectural views or documents, the model guides practitioners in determining what architectural knowledge should be produced, when, and for whom, complementing existing view-based architecture description standards (ISO/IEC/IEEE 42010, 2011). The ADM is designed to integrate with the software lifecycle and organizational governance mechanisms, enabling traceability between project characteristics, architectural concerns, and decision outcomes across system evolution (Bosch, 2014; Lago et al., 2015). A case study illustrates how the model supports consistent architectural decision making over time while accommodating changing project constraints.

1. Introduction

Architecture documentation frameworks based on multiple views have been widely adopted to structure architectural descriptions and improve stakeholder communication (Kruchten, 1995; ISO/IEC/IEEE 42010, 2011). These frameworks provide clear guidance on how architectural information may be organized once relevant views are identified (Rozanski and Woods, 2012). However, decisions regarding the scope, timing, and level of architectural documentation are often left implicit, relying heavily on individual experience and local practices (van Heesch et al., 2012). As a result, projects with different characteristics may apply similar documentation structures without an explicit rationale linking project context to architectural decision making (Waterman et al., 2015).

In contrast, industrial experience increasingly demonstrates that architecture effort should be selectively allocated based on actual project characteristics, such as integration complexity, risk exposure, and operational criticality (Bosch, 2014; Lago et al., 2015). In many projects, the challenge is not how to describe architecture using views, but which architectural information must be made explicit at all to support sound decision making and effective review (Hilliard

et al., 2020).

This paper addresses this challenge by focusing on architecture artifact decisions rather than architectural design decisions. We propose an Architecture Artifact Decision Model (AADM) that determines which architecture artifacts are required, recommended, or optional for a given project, based on its dominant complexity and risk characteristics. Architectural viewpoints are considered only after artifact decisions have been made and are used to structure and present selected artifacts, aligning with the separation between concerns, viewpoints, and views defined in existing architecture description standards (ISO/IEC/IEEE 42010, 2011).

By shifting the decision focus from which viewpoints to apply to which artifacts to produce, the proposed approach emphasizes risk mitigation, decision relevance, and review efficiency, rather than documentation completeness or exhaustive architectural description.

2. Research Scope and Positioning

2.1 Scope

This work focuses on decision guidance for selecting architecture artifacts in the context of software architecture design and review. The scope is explicitly limited to decisions about what architectural artifacts should be produced, not how those artifacts are defined or governed, reflecting prior observations that architecture decision support often operates at different abstraction levels within software engineering processes (van Heesch et al., 2012; Hilliard et al., 2020).

Specifically, this paper:

- Provides project-type–driven guidance for selecting architecture artifacts, in line with calls for context-sensitive and risk-aware architecture practices (Bosch, 2014; Waterman et al., 2015)
- Defines when specific artifacts are mandatory, recommended, or optional, supporting proportional allocation of architecture effort based on project needs (Lago et al., 2015)
- Bases artifact selection on system complexity and risk profile, consistent with risk-informed approaches to architectural decision making (Kruchten et al., 2019)
- Treats architectural viewpoints as realization mechanisms, not as decision drivers, aligning with the conceptual separation between concerns, viewpoints, and views established in architecture description standards (ISO/IEC/IEEE 42010, 2011)

2.2 Out of Scope

This paper does not:

- Define architecture artifacts, viewpoints, or modeling notations
- Prescribe governance processes, approval workflows, or compliance mechanisms
- Recommend specific technologies, platforms, or implementations

These aspects are assumed to be addressed by existing Enterprise Architecture standards, viewpoint frameworks, and governance guides, which provide the necessary definitions and controls once artifact decisions have been made (ISO/IEC/IEEE 42010, 2011; Rozanski and Woods, 2012).

3. Abstraction and Core Principles

3.1 Project Characteristics as the Primary Decision Abstraction

The proposed model adopts project characteristics as the primary abstraction for architecture artifact decisions. Rather than classifying systems by technology labels, decisions are driven by characteristics that directly influence architectural risk and complexity, consistent with prior research emphasizing risk- and context-driven architectural reasoning over technology-centric classification (Bosch, 2014; Kruchten et al., 2019). Relevant project characteristics include:

- Integration breadth and depth
- Identity and trust boundaries
- Runtime and interaction complexity
- Data sensitivity and regulatory exposure
- Operational, performance, and resilience risks

These characteristics have been widely recognized as dominant drivers of architectural concerns and long-term system evolution in software-intensive systems (Lago et al., 2015; Hilliard et al., 2020). Technology labels such as *Web*, *AI*, or *Data* are therefore treated as secondary indicators that may correlate with certain risks, but do not determine artifact selection by themselves, reflecting observations from industrial and empirical studies on architecture decision making (Waterman et al., 2015).

3.2 Separation of Viewpoint Decisions and Artifact Decisions

This work explicitly separates two types of architecture-related decisions:

- Viewpoint decisions, which determine which architectural perspectives are relevant for a project
- Artifact decisions, which determine which concrete architectural artifacts must be produced

The Architecture Artifact Decision Model (AADM) addresses the latter. Viewpoints are assumed to be relatively stable and reusable across projects, whereas artifact selection is project-specific, dynamic, and risk-driven. This separation aligns with the conceptual distinction between concerns, viewpoints, and views established in architecture description standards (ISO/IEC/IEEE 42010, 2011), while addressing the practical need for differentiated documentation effort observed in evolving and heterogeneous project environments (Bosch, 2014).

By decoupling artifact selection from viewpoint selection, the proposed approach allows architecture documentation to remain flexible and efficient without undermining architectural rigor, a concern frequently raised in studies of architecture practices in agile and large-scale development contexts (Waterman et al., 2015).

3.3 Artifact Decisions as Architecture Decisions

Although the proposed model does not address architectural design choices such as technology selection or structural styles, artifact selection decisions are treated as architecture-related decisions because they determine:

- Which architectural concerns become explicit
- Which risks are surfaced and reviewed
- Which trade-offs are made visible to stakeholders

Prior work on architectural knowledge management and decision rationale has shown that such meta-decisions strongly influence how architectural knowledge is produced, communicated, and evaluated over time (van Heesch et al., 2012; Hilliard et al., 2020). In this sense, artifact decisions act as higher-level decisions that shape architectural reasoning and review outcomes, even in the absence of direct design prescriptions.

4. AADM Decision Logic: From Project to Architecture Artifacts

4.1 Overview of the Decision Logic

The Architecture Artifact Decision Model (AADM) provides a structured decision logic for determining which architecture artifacts should be produced for a given project. Rather than optimizing documentation completeness or prescribing a fixed artifact set, the model focuses on making essential architectural concerns explicit, based on project-specific complexity and risk characteristics, reflecting prior observations that architecture documentation should be decision-driven rather than checklist-based (Waterman et al., 2015; Bosch, 2014).

The core assumption underlying AADM is that architecture artifacts are justified only insofar as they expose relevant architectural concerns and risks. Consequently, artifact selection is treated as a decision problem, rather than as a documentation task, consistent with research emphasizing architecture documentation as a form of decision support (van Heesch et al., 2012; Hilliard et al., 2020).

The AADM decision logic follows a three-stage reasoning process:

1. Project characterization
2. Risk and concern inference
3. Artifact selection and classification

This process establishes a clear separation between decision rationale and artifact realization, allowing architectural viewpoints to be applied only after artifact decisions have been made, in line with the conceptual separation between concerns, viewpoints, and views defined in architecture description standards (ISO/IEC/IEEE 42010, 2011).

4.2 Project Characterization

The first step of AADM is to characterize the project in terms of dominant architectural drivers, rather than implementation technologies or delivery methodologies. This approach aligns with prior work advocating context- and risk-driven architectural reasoning over technology-centric classification (Bosch, 2014; Kruchten et al., 2019).

Project characterization focuses on identifying characteristics that have a direct impact on architectural risk exposure and review needs, including but not limited to:

- **Integration characteristics**
  Number of interacting systems, degree of coupling, and cross-domain dependencies
- **Identity and trust boundaries**
  Authentication mechanisms, authorization scope, tenant separation, and partner

integration

- **Runtime and interaction complexity**
  Asynchronous behavior, stateful interactions, long-running processes, and coordination logic
- **Data characteristics**
  Data sensitivity, compliance requirements, lineage, and movement across system boundaries
- **Operational criticality**
  Availability expectations, performance constraints, resilience requirements, and failure impact

These characteristics correspond to widely recognized drivers of architectural risk and long-term system evolution (Lago et al., 2015; Hilliard et al., 2020). They are assessed qualitatively and are intended to support comparative reasoning rather than precise measurement, consistent with common practice in architecture evaluation and review (Kruchten et al., 2019).

4.3 Risk-Driven Inference of Architectural Concerns

Based on the identified project characteristics, AADM infers a set of architectural concerns that must be explicitly addressed during design and review. The model assumes that architectural risks often remain implicit unless they are made visible through appropriate architectural representations, a recurring observation in studies of architecture decision making and review practices (van Heesch et al., 2012; Waterman et al., 2015).

Typical concern categories include:

- Misalignment or ambiguity in system interactions
- Inadequate handling of identity propagation and access control
- Unclear runtime behavior under load or failure
- Insufficient visibility into data movement and compliance exposure
- Weak understanding of operational dependencies and failure modes

AADM does not attempt to enumerate all possible concerns. Instead, it provides a reasoning framework for identifying which concerns are significant enough to justify explicit architectural treatment in a given project context, consistent with risk-informed approaches to architectural analysis (Bosch, 2014; Lago et al., 2015).

4.4 Artifact Selection as a Decision Outcome

Once dominant concerns have been identified, AADM determines which architecture artifacts are needed to make those concerns explicit and reviewable. Treating artifact selection as an explicit decision outcome aligns with prior work emphasizing proportional and purpose-driven architecture documentation (Waterman et al., 2015).

Artifact selection is guided by the following principles:

- **Necessity**
  An artifact is selected only if it contributes to exposing or mitigating a relevant concern
- **Sufficiency**
  Artifacts should collectively cover the identified concerns without unnecessary overlap
- **Proportionality**

The level of architectural effort should be proportional to project complexity and risk Artifacts are then classified into three categories:

- **Mandatory**: Artifacts required to support basic architectural understanding and risk assessment
- **Recommended**: Artifacts that address common or elevated risks but may be omitted with justification
- **Optional**: Artifacts that primarily support communication or understanding in non-critical areas

This classification expresses decision strength rather than artifact importance or quality, supporting transparent and explainable architecture review practices (Hilliard et al., 2020).

4.5 Project-Type–Driven Decision Baselines

To support consistent decision making across projects, AADM provides project-type–driven baselines that map common project characteristics to default artifact selections. Similar baseline-oriented approaches have been observed in industrial architecture practices, although often without explicit risk rationale (Waterman et al., 2015).

Each project is mapped to the closest project type based on its dominant complexity and risk profile, rather than its technology stack. These baselines serve as starting points rather than rigid prescriptions, reflecting the need for adaptable architecture processes in evolving project environments (Bosch, 2014).

Projects may tailor artifact selection by:

- Reclassifying recommended artifacts as optional
- Elevating optional artifacts to recommended or mandatory
- Substituting artifacts with equivalent representations

Such tailoring is considered acceptable provided that all mandatory concerns are addressed and the decision rationale is made explicit, consistent with governance principles emphasizing justification over compliance (Hilliard et al., 2020).

4.6 Relationship to Architectural Viewpoints

AADM deliberately decouples artifact selection from viewpoint selection. Architectural viewpoints are assumed to be stable and reusable perspectives that can be applied to structure selected artifacts, as defined in established architecture description standards (ISO/IEC/IEEE 42010, 2011).

In the proposed approach:

- Artifacts determine what architectural information must be produced
- Viewpoints determine how that information is organized and presented

A single artifact may be realized through one or multiple viewpoints, depending on project needs. This flexibility allows AADM to support both lightweight and comprehensive documentation practices without compromising decision transparency, addressing concerns raised in studies of architecture documentation scalability (Waterman et al., 2015).

4.7 Decision Transparency and Traceability

Although AADM does not prescribe specific documentation formats, it requires that artifact decisions be traceable to project characteristics and inferred concerns. Traceability between

context, concerns, and architectural decisions has been widely recognized as essential for effective architecture review and long-term system evolution (van Heesch et al., 2012; Hilliard et al., 2020).

At a minimum, each artifact decision should be explainable in terms of:

- Which project characteristics motivated the decision
- Which architectural concerns are addressed
- Why the selected artifact category (mandatory, recommended, optional) is appropriate

This level of transparency is sufficient to support architecture review and cross-team communication without imposing additional governance overhead.

4.8 Summary

The AADM decision logic provides a structured, risk-aware approach for determining which architecture artifacts are worth producing in a given project context. By treating artifact selection as a first-class architectural decision and deferring viewpoint realization, the model enables architecture practices that are focused, adaptive, and economically justified, aligning with established principles of continuous and evolution-aware software architecture (Bosch, 2014; Lago et al., 2015).

5. Project-Type–Driven Artifact Matrix

5.1 Purpose of the Artifact Matrix

The Project-Type–Driven Artifact Matrix provides a practical decision baseline for applying the Architecture Artifact Decision Model (AADM) across heterogeneous projects. Its purpose is not to prescribe a fixed documentation template, but to operationalize artifact decisions by mapping recurring project characteristics to typical architectural risks and corresponding artifact needs. This operationalization responds to long-standing observations that architecture practices often lack explicit decision rationale linking project context, risk exposure, and documentation effort (Waterman et al., 2015; Bosch, 2014).

The matrix addresses three recurring challenges observed in industrial practice:

1. Architecture effort is often allocated uniformly across projects with vastly different risk profiles.
2. Architecture reviews tend to focus on document completeness rather than decision relevance.
3. Teams lack a shared reference for justifying why certain artifacts are produced or omitted.

Similar concerns have been reported in empirical studies of architecture documentation and review practices, particularly in large-scale and multi-team environments (Hilliard et al., 2020; van Heesch et al., 2012). By making artifact expectations explicit at the project-type level, the matrix supports consistent yet flexible decision making.

5.2 Project Type Classification

Each project is mapped to a project type based on its dominant complexity and risk characteristics, rather than its implementation technology. Project types are therefore analytical categories, not organizational or technical labels. This aligns with prior work arguing that architectural risk and decision needs are driven primarily by system characteristics and

operational context, rather than by technology stacks alone (Bosch, 2014; Kruchten et al., 2019).

The project types covered in the matrix include:

- Web applications and service-oriented systems
- Data processing systems (ETL / ELT / BI)
- AI systems (ML / LLM / RAG / MLOps / LLMOps)
- Complex business process systems
- Identity- and interaction-heavy integration systems
- Performance- and resilience-critical integration systems
- Systems with complex logic, such as workflows, rule engines, and transaction control

A project may exhibit characteristics of multiple types; in such cases, the matrix is applied using the most risk-dominant type as the baseline, reflecting common practice in risk-driven architecture evaluation (Lago et al., 2015).

5.3 Artifact Classification Semantics

For each project type, the matrix classifies architecture artifacts into three categories:

- **Mandatory (✓)**
  Artifacts required to support essential architectural understanding and risk assessment. Mandatory does not imply exhaustive design documentation.
- **Recommended (R)**
  Artifacts that typically expose elevated or non-obvious risks, especially related to runtime behavior, non-functional properties, or cross-system interactions.
- **Optional (○)**
  Artifacts that primarily support communication, clarification, or non-critical design areas. Optional artifacts must not become governance gating items.

This classification expresses decision strength rather than artifact importance or quality, supporting proportional documentation practices advocated in architecture process literature (Waterman et al., 2015; Hilliard et al., 2020).

5.4 Structure of the Artifact Matrix

The matrix is organized by project type (rows) and architecture artifacts (columns). Each cell indicates the default decision strength for producing a given artifact in a given project type.

The artifact set referenced in the matrix includes commonly used architecture artifacts such as:

- Technical Diagram
- Application Collaboration Diagram
- Business Diagram
- Data Model
- Authentication and Service Flow
- Business Process Diagram
- Data Pipeline Diagram
- System Process Flow

These artifacts are assumed to be defined and standardized elsewhere within Enterprise Architecture practices and viewpoint frameworks (ISO/IEC/IEEE 42010, 2011; Rozanski and Woods, 2012).

5.5 Matrix Interpretation by Project Type

Rather than interpreting the matrix mechanically, AADM requires understanding the risk rationale behind each classification. This interpretive stance is consistent with prior findings that architecture documentation is most effective when used as a decision support instrument rather than a compliance checklist (van Heesch et al., 2012).

Web Applications and Service-Oriented Systems

Web-oriented projects typically involve moderate integration complexity and well-understood interaction patterns. As a result:

- Structural artifacts (e.g., Technical Diagram, Application Collaboration Diagram) are mandatory.
- Interaction, data, and process-oriented artifacts are generally optional unless elevated risks are present.

This reflects the relatively predictable runtime and operational behavior of typical web systems (Bosch, 2014).

Data Projects (ETL / ELT / BI)

Data projects are characterized by data movement, transformation, and compliance exposure:

- Data Pipeline and related flow artifacts are mandatory to expose lineage and processing logic.
- Structural and collaboration artifacts remain necessary to understand system boundaries.
- Business and interaction diagrams are often optional unless driven by regulatory or organizational concerns.

These characteristics are widely recognized as dominant risk drivers in data-intensive systems (Lago et al., 2015).

AI Projects (ML / LLM / RAG / MLOps / LLMOps)

AI projects introduce risks related to model lifecycle, data dependency, and operational behavior:

- Structural and collaboration artifacts are mandatory to clarify system integration.
- Data pipeline artifacts are typically mandatory or recommended due to training and inference flows.
- System process flow artifacts are recommended to expose runtime orchestration and feedback loops.

This aligns with emerging observations on the architectural complexity and operational uncertainty of AI-enabled systems (Bosch, 2014; Kruchten et al., 2019).

Complex Business Process Systems

Systems dominated by long-running, stateful, or highly coordinated business processes require explicit treatment of control flow and responsibility boundaries:

- Business Process artifacts are mandatory.
- Structural and collaboration artifacts support understanding of process ownership and integration.
- Runtime and data artifacts are selected based on process criticality.

Such systems have been shown to benefit from explicit process-centric architectural representations (Rozanski and Woods, 2012).

Identity- and Interaction-Heavy Integration Systems

These systems are dominated by authentication, authorization, and trust propagation risks, such as SSO, federated identity, and partner integration:

- Authentication and Service Flow artifacts are mandatory.
- Structural and collaboration artifacts are required to expose identity boundaries.
- Runtime artifacts are recommended when multi-hop or cross-domain interactions are present.

Identity-related concerns are a well-documented source of architectural risk in distributed systems (Hilliard et al., 2020).

Performance- and Resilience-Critical Integration Systems

Projects with strict latency, throughput, or availability requirements exhibit elevated operational risks:

- System Process Flow artifacts are mandatory to expose runtime behavior under load.
- Structural and collaboration artifacts remain mandatory.
- Data and business artifacts are selected based on failure impact and recovery strategies.

This reflects established practices in architecture evaluation for performance-critical systems (Lago et al., 2015).

Systems with Complex Logic

Systems implementing workflows, rule engines, or transaction coordination require visibility into decision logic and state transitions:

- System Process Flow artifacts are mandatory.
- Structural artifacts provide necessary context.
- Other artifacts are selected based on integration and operational exposure.

5.6 Tailoring and Deviation

The matrix represents a default decision baseline rather than a rigid prescription. Projects may tailor artifact selection by:

- Omitting recommended artifacts with explicit justification
- Elevating optional artifacts when local risks warrant it
- Substituting artifacts with equivalent representations

Such deviations are considered acceptable provided that mandatory concerns are addressed and the decision rationale is documented, consistent with governance approaches emphasizing accountability over uniform compliance (Hilliard et al., 2020).

5.7 Relationship to Governance and Review

The artifact matrix supports architecture review by focusing attention on decision-relevant artifacts rather than enforcing documentation completeness. Mandatory artifacts form the minimum basis for architecture review, while recommended artifacts highlight areas where

hidden or non-functional risks are likely to emerge.
The matrix itself does not impose governance rules; instead, it provides decision transparency that governance processes may build upon, aligning with the separation between decision support and governance enforcement advocated in architecture process literature (ISO/IEC/IEEE 42010, 2011).

5.8 Summary

The Project-Type–Driven Artifact Matrix operationalizes AADM by linking project characteristics, typical risk profiles, and artifact decisions. By providing a shared baseline that is both risk-aware and tailorable, the matrix enables architecture practices that are consistent, efficient, and aligned with actual project needs, supporting evolution-aware and process-centric software architecture management (Bosch, 2014; Waterman et al., 2015).

6. Recommended Usage and Lifecycle Integration

6.1 Intended Usage of AADM

The Architecture Artifact Decision Model (AADM) and the Project-Type–Driven Artifact Matrix are intended to be used as decision-support instruments rather than as mandatory process steps or documentation templates. This positioning reflects prior observations that architecture methods are most effective when they support reasoning and decision making, rather than enforcing prescriptive workflows (van Heesch et al., 2012; Hilliard et al., 2020).
Their primary purpose is to help architects and project teams:

- Identify which architectural information must be made explicit
- Allocate architecture effort proportionally to project risk and complexity
- Focus architecture review on critical decisions rather than document volume

The model assumes iterative and lightweight application, consistent with modern agile and incremental delivery practices, where architectural decisions and documentation evolve alongside the system (Bosch, 2014; Waterman et al., 2015).

6.2 Usage During Solution Design

During early solution design, AADM can be applied to establish an architecture documentation baseline. Similar baseline-oriented approaches have been observed in industrial architecture practices, particularly in environments with heterogeneous project portfolios (Waterman et al., 2015).
Typical usage includes:

1. Characterizing the project based on dominant complexity and risk drivers
2. Mapping the project to the closest project type in the artifact matrix
3. Identifying mandatory and recommended artifacts as the initial baseline
4. Tailoring the baseline where project-specific risks justify deviation

At this stage, the goal is not to fully specify architecture artifacts, but to clarify expectations regarding which architectural concerns must be addressed. This aligns with the principle of delaying unnecessary design detail while preserving decision relevance (Bosch, 2014).

6.3 Usage During Architecture Review

During architecture review, the artifact matrix supports a deliberate shift in review focus. Rather than assessing documentation completeness, reviewers can:

- Verify that all mandatory artifacts are present
- Examine whether selected artifacts adequately expose identified risks
- Evaluate trade-offs and rationale captured within those artifacts

Recommended artifacts act as risk indicators, guiding reviewers toward areas where deeper discussion may be warranted. This usage aligns with established views that architecture reviews should prioritize risk mitigation and decision quality over formal completeness (Lago et al., 2015; Hilliard et al., 2020).

This approach enables more effective and time-efficient reviews, particularly in environments with high project throughput.

6.4 Usage in Agile and Incremental Delivery

AADM is designed to support incremental and evolving architectures. Artifact decisions may be revisited as projects evolve, for example when:

- Integration scope increases
- Operational criticality changes
- New regulatory or security requirements emerge

In such cases, artifact selection can be adjusted by reapplying the matrix without invalidating previously produced artifacts. This supports continuous alignment between architecture effort and project reality, a key requirement in long-lived and continuously evolving software systems (Bosch, 2014; Kruchten et al., 2019).

6.5 Tailoring and Justification

Tailoring is an expected and legitimate part of using AADM. Projects may:

- Omit recommended artifacts with explicit justification
- Produce optional artifacts to address local communication needs
- Replace artifacts with equivalent representations

The key requirement is that tailoring decisions remain explainable in terms of project characteristics and risk considerations. This justification-oriented approach reflects governance principles emphasizing accountability and transparency over uniform compliance (Hilliard et al., 2020).

6.6 Integration with Governance Processes

Although AADM does not prescribe governance processes, it is compatible with common Enterprise Architecture governance practices. In governance contexts, the artifact matrix can be used to:

- Define minimum expectations for architecture review readiness
- Provide a transparent rationale for documentation scope
- Reduce subjective disagreement about “how much architecture is enough”

Such use aligns with the separation between decision support and governance enforcement established in architecture description standards and governance literature (ISO/IEC/IEEE 42010, 2011; Rozanski and Woods, 2012). Governance teams may adopt the matrix as a

reference while allowing projects to retain ownership of tailoring decisions.

6.7 Roles and Responsibilities

The model does not mandate specific roles, but typical responsibilities include:

- **Architects**: applying AADM to determine artifact needs and explain decisions
- **Project teams**: producing selected artifacts at an appropriate level of detail
- **Reviewers**: focusing on risks and decisions exposed by mandatory artifacts

This role-agnostic design supports adoption across diverse organizational structures and aligns with observations that effective architecture practices must adapt to local team configurations (Waterman et al., 2015).

6.8 Summary

By integrating AADM and the Project-Type–Driven Artifact Matrix into solution design, architecture review, and iterative delivery, organizations can achieve risk-aware, proportional, and efficient architecture documentation practices. The recommended usage emphasizes decision transparency and adaptability, enabling architecture effort to evolve with project needs rather than being fixed upfront, consistent with process- and evolution-oriented views of software architecture (Bosch, 2014; Lago et al., 2015).

7. Discussion: Artifact-First versus View-First Architecture Documentation

7.1 View-First Approaches and Their Assumptions

View-based architecture documentation frameworks assume that a predefined set of architectural viewpoints can be applied uniformly across systems. In such approaches, the primary decision is which viewpoints to adopt, and architecture artifacts are subsequently produced as realizations of those viewpoints. This logic is well established in architecture description standards and viewpoint-oriented frameworks, which emphasize consistency, comparability, and stakeholder communication (ISO/IEC/IEEE 42010, 2011; Rozanski and Woods, 2012).

This view-first logic rests on several implicit assumptions:

- Architectural viewpoints are equally relevant across projects
- Documentation effort scales linearly with the number of viewpoints
- Producing a view inherently ensures that relevant architectural concerns are addressed

While these assumptions support standardization, prior work has noted that they are increasingly strained in environments characterized by heterogeneous systems, varying risk profiles, and agile or incremental delivery constraints (Bosch, 2014; Waterman et al., 2015).

7.2 Limitations of View-First Documentation in Practice

Industrial practice reveals several recurring limitations of view-first approaches, particularly when documentation scope is driven primarily by fixed viewpoint sets rather than project context.

1. **Over-documentation**

Projects are often required to produce views that provide limited decision value relative to their effort, a phenomenon frequently reported in empirical studies of architecture documentation practices (van Heesch et al., 2012).

2. **Weak Risk Alignment**
   View selection does not necessarily correlate with dominant project risks, leading to blind spots in security, operational, or integration concerns. Risk-driven architecture evaluation studies have repeatedly emphasized the need to make such concerns explicit rather than assuming coverage through representation alone (Lago et al., 2015).
3. **Review Inefficiency**
   Architecture reviews tend to focus on whether all views are present, rather than whether critical decisions and trade-offs are sufficiently exposed. This shift from decision quality to documentation completeness has been identified as a recurring challenge in large-scale review settings (Hilliard et al., 2020).
4. **Limited Adaptability**
   View sets are typically static, making it difficult to adapt documentation scope as project complexity evolves. This limitation conflicts with observations from agile and continuous delivery environments, where architectural focus must shift over time (Bosch, 2014).

These limitations are not inherent flaws of viewpoints themselves, but consequences of treating viewpoint selection as the primary decision mechanism.

7.3 Artifact-First Decision Logic

The artifact-first approach proposed in this paper inverts the traditional logic:

- Architecture artifacts are selected first, based on project-specific risk and complexity
- Architectural viewpoints are applied second, as realization mechanisms

In this approach, the key decision is not how to look at the architecture, but which architectural information must be made explicit to support risk-aware decision making. This perspective aligns with research on architectural knowledge management, which emphasizes that making the right information explicit is a prerequisite for effective reasoning and review (van Heesch et al., 2012; Hilliard et al., 2020).

By focusing on artifact selection, the model directly addresses the question of documentation necessity, which is often left implicit in view-first frameworks.

7.4 Complementarity Rather Than Replacement

Artifact-first and view-first approaches are not mutually exclusive. Viewpoints remain essential for structuring architectural information and supporting stakeholder communication, as consistently emphasized in architecture description literature (ISO/IEC/IEEE 42010, 2011; Rozanski and Woods, 2012).

In the proposed model, viewpoints are deliberately decoupled from artifact decisions to avoid conflating representation choices with decision necessity. In this sense, the artifact-first approach complements existing view-based frameworks by introducing a prior decision layer that governs documentation scope and effort, rather than attempting to replace established viewpoint concepts.

7.5 Implications for Architecture Review and Governance

Adopting an artifact-first decision logic has several implications:

- Architecture reviews shift from completeness checks to risk-focused evaluation
- Governance discussions become more transparent, as documentation expectations are justified by project characteristics
- Architectural consistency is maintained at the decision level rather than at the artifact count level

These implications align with calls for more proportional and context-sensitive governance in fast-moving and diverse system landscapes, where uniform documentation rules are increasingly difficult to justify (Hilliard et al., 2020; Bosch, 2014).

7.6 Methodological Implications

From a methodological perspective, this work suggests that architecture documentation should be understood as a decision-support activity rather than a static deliverable. By explicitly modeling artifact decisions, the proposed approach:

- Makes architectural reasoning more transparent
- Reduces accidental complexity in documentation practices
- Enables more economically justified allocation of architecture effort

This perspective extends existing architecture documentation research by shifting attention from representational structures to meta-decisions that shape how architectural knowledge is produced, communicated, and evaluated (van Heesch et al., 2012; Waterman et al., 2015).

7.7 Limitations and Threats to Validity

The proposed model is conceptual and prescriptive in nature. It does not claim empirical optimality or quantitative risk assessment accuracy. Potential limitations include:

- Dependence on qualitative judgment when characterizing projects
- Variability in how organizations define and standardize artifacts
- The need for organizational discipline to maintain decision rationale transparency

These limitations reflect common challenges in architecture practice and documentation governance and do not negate the model's intended applicability (Hilliard et al., 2020).

7.8 Summary

The artifact-first approach presented in this paper challenges the assumption that architectural viewpoints should be the primary driver of documentation decisions. By treating artifact selection as a first-class, risk-aware architectural decision, the proposed model provides a flexible and review-focused alternative that complements existing view-based frameworks rather than replacing them.

8. Related Work

This section positions the proposed Architecture Artifact Decision Model (AADM) with respect to existing work on architecture decisions, view-based documentation standards, risk-aware

decision frameworks, and industrial artifact selection practices.

8.1 Architecture Decision Records (ADR)

Architecture Decision Records (ADR) are a widely adopted practice for capturing individual architectural decisions and their rationale. ADRs focus on documenting what decision was made, why it was made, and what alternatives were considered (Nygard, 2011; van Heesch et al., 2012).

ADRs operate at the level of individual decision instances, rather than at the level of documentation strategy. As such, they do not provide guidance on:

- Which architectural concerns should be documented
- Which artifacts are necessary for a given project
- How documentation scope should be adjusted based on project risk

AADM complements ADRs by addressing a meta-decision layer: determining which architectural information must be made explicit before individual decisions are recorded. In this sense, ADRs can be understood as potential contents of selected artifacts, rather than substitutes for artifact selection decisions.

8.2 AVDM: A Risk-Informed Decision Model for Architecture Viewpoint Selection

Recent work has proposed decision-oriented models that treat architecture documentation choices as explicit architectural decisions rather than as fixed documentation tasks. A representative example is the Architecture Viewpoint Decision Model (AVDM), which addresses the problem of how architectural viewpoints should be selected and instantiated in specific project contexts (Han, 2024).

AVDM is motivated by the observation that architectural documentation is often driven by predefined checklists or conventions, leading either to excessive artifacts with limited decision value or to insufficient visibility into dominant architectural uncertainty. While existing viewpoint taxonomies clarify available viewpoints and associated concerns, they provide limited guidance on how viewpoints should be prioritized and applied based on project-specific risks (Rozanski and Woods, 2012; ISO/IEC/IEEE 42010, 2011).

To address this gap, AVDM treats viewpoint selection as a risk-informed decision activity. Architectural risk is used as an informational signal to prioritize viewpoints, and the expected decision value of viewpoint instantiation is evaluated based on project characteristics. Rather than prescribing fixed documentation sets, AVDM classifies viewpoint instantiation outcomes as mandatory, recommended, or optional, enabling proportional and context-sensitive documentation practices.

Multiple industrial case studies reported in the AVDM work illustrate how risk-informed viewpoint selection can improve decision transparency, focus architecture reviews, and expose cross-domain and non-functional uncertainty (Han, 2024). Importantly, AVDM positions viewpoint decision making as a structuring mechanism for architectural judgment, rather than as a replacement for professional expertise.

From a positioning perspective, AVDM and the proposed Architecture Artifact Decision Model (AADM) operate at complementary decision layers. AVDM focuses on determining which architectural viewpoints are worth instantiating to support design reasoning and review, whereas AADM addresses a prior meta-decision: determining which architecture artifacts

should be produced at all, based on project characteristics and risk profiles. In this sense, AADM provides decision guidance at the documentation scope level, while AVDM provides guidance at the viewpoint selection level within that scope.

8.3 View-Based Architecture Documentation Standards (ISO/IEC/IEEE 42010)

ISO/IEC/IEEE 42010 defines a conceptual framework for architecture descriptions based on stakeholders, concerns, viewpoints, and views. It establishes a common vocabulary and structure for organizing architectural information (ISO/IEC/IEEE 42010, 2011).

The standard intentionally avoids prescribing:

- Which viewpoints must be used
- How many views should be produced
- How documentation effort should scale with project characteristics

AADM aligns with ISO 42010 by respecting the separation between concerns, viewpoints, and views, but extends the framework by providing decision guidance prior to viewpoint selection. In the proposed approach, viewpoints are treated as realization mechanisms chosen after artifact decisions have been made, thereby preserving compatibility with ISO 42010 while addressing a gap in practical decision guidance noted in prior empirical studies (Waterman et al., 2015).

8.4 Industrial Artifact Matrices and Reference Checklists

Many organizations employ internal architecture artifact matrices, checklists, or reference templates to standardize documentation expectations across projects. Such practices have been widely reported in industrial experience studies and practitioner literature, reflecting recognition of the need for differentiated documentation effort (Bosch, 2014; Hilliard et al., 2020).

However, these practices often exhibit limitations:

- Project categories are coarse and technology-centric
- Risk considerations are implicit rather than explicit
- Tailoring decisions lack formal justification mechanisms

AADM generalizes and formalizes this industrial intuition by making project characteristics and risk profiles explicit decision inputs, and by framing artifact selection as an explainable architectural decision rather than an administrative rule.

8.5 Positioning Summary

Table 2 summarizes the positioning of AADM relative to related work.

Approach	Primary Focus	Decision Level	Addresses Artifact Selection
ADR	Individual design decisions	Micro (decision instance)	No
AVDM	Risk-informed architecture viewpoint selection	Meso (viewpoint decision process)	No
ISO/IEC/IEEE 42010	Architecture description structure	Conceptual framework	No

Approach	Primary Focus	Decision Level	Addresses Artifact Selection
Industrial matrices	Documentation standardization	Administrative	Implicit
AADM (this work)	Artifact decision guidance	Meta (documentation scope)	**Yes**

8.6 Summary

Existing architecture decision and documentation approaches provide valuable support at the levels of decision capture, risk evaluation, and structural organization. This work contributes at the process-decision level rather than at the technical system level, focusing on how documentation scope and architectural artifacts evolve across projects and lifecycles (Bosch, 2014; Hilliard et al., 2020).

AADM addresses this gap by introducing an explicit decision model for artifact selection that complements, rather than replaces, existing standards and practices.

9. Conclusion and Future Work

This paper addressed a recurring problem in software architecture practice: the lack of systematic, risk-aware guidance for deciding how much architectural documentation is necessary for a given project. Empirical and industrial studies have repeatedly observed that architecture documentation effort is often misaligned with actual decision needs, particularly in heterogeneous and rapidly evolving system landscapes (Bosch, 2014; Waterman et al., 2015).

Traditional view-based approaches emphasize standardization and structural consistency, but frequently leave the scope of architecture artifacts implicit and detached from project-specific risk and complexity. As a result, documentation practices tend to focus on completeness rather than on decision relevance and risk exposure (Hilliard et al., 2020).

To address this gap, this paper proposed the Architecture Artifact Decision Model (AADM), a project-type–driven decision model that determines which architecture artifacts are mandatory, recommended, or optional based on project characteristics and risk exposure. Rather than prescribing architectural views or documentation formats, AADM treats architectural views as realization mechanisms applied after artifact decisions have been made, remaining compatible with existing view-based standards (ISO/IEC/IEEE 42010, 2011).

Contributions

The main contributions of this work are:

1. **An artifact-first decision perspective**
   This paper reframes architecture documentation as a decision-support activity and elevates artifact selection to a first-class architectural decision that precedes viewpoint realization. This perspective builds on prior work on architectural decision making and knowledge management while shifting the focus to documentation

scope decisions that are typically left implicit (van Heesch et al., 2012; Hilliard et al., 2020).

2. **A project-type–driven artifact decision model**
   AADM provides explicit guidance on how project characteristics, system complexity, and risk profiles influence artifact selection, enabling proportional allocation of architecture effort across diverse project contexts (Bosch, 2014).
3. **A structured artifact matrix for practical adoption**
   The Project-Type–Driven Artifact Matrix offers a lightweight and explainable baseline that supports tailoring while preserving review consistency, reflecting industrial practices while making their underlying decision logic explicit (Waterman et al., 2015).
4. **Positioning within existing standards and practices**
   The proposed approach complements view-based standards such as ISO/IEC/IEEE 42010 and risk-aware decision frameworks by addressing a distinct meta-decision layer that has received limited explicit treatment in prior work.

Practical Implications

By making artifact decisions explicit and justifiable, AADM supports:

- More focused and efficient architecture reviews
- Reduced documentation overhead in low-risk projects
- Improved transparency in governance discussions
- Better alignment between architecture effort and system risk

The model is intentionally lightweight and compatible with agile and incremental delivery models, where architectural focus must evolve over time rather than being fixed upfront (Bosch, 2014).

Limitations

This work is conceptual and prescriptive in nature. It does not claim empirical optimality, nor does it provide quantitative risk assessment or cost–benefit models. The effectiveness of AADM depends on informed judgment when characterizing projects and on organizational discipline in maintaining decision rationale transparency. These limitations are consistent with known challenges in architecture documentation and governance practices (Hilliard et al., 2020).

Future Work

Several directions for future research are identified:

- **Empirical evaluation of AADM in industrial settings**, including qualitative studies on review efficiency, decision transparency, and perceived documentation value
- **Integration with architecture decision records**, exploring how artifact-level decisions and individual decision instances can be systematically linked (van Heesch et al., 2012)
- **Tool support**, enabling automated or semi-automated artifact recommendations based on project characteristics
- **Refinement of risk characterization**, potentially incorporating quantitative indicators or historical project data

These directions aim to strengthen the empirical grounding and operational scalability of the proposed approach without altering its decision-oriented nature.

Closing Remarks

By explicitly modeling the decision logic that governs architecture artifact selection, this paper contributes a missing link between architectural theory and everyday documentation practice. The proposed artifact-first approach offers a pragmatic complement to view-driven documentation, supporting risk-aware, proportional, and explainable architecture decisions across system lifecycles and organizational contexts.

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